

Origin of the CEEX–PHOKHARA discrepancy in the pion angular distributions

KLOE large-angle scenario, $e^+e^- \rightarrow \pi^+\pi^-\gamma$ at NLO, $F_\pi = 1$, no vacuum polarisation

Jeremy Paltrinieri*

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Abstract

In the KLOE large-angle (KLOE-LA) scenario the NLO predictions of CEEX and PHOKHARA 10.0 for $e^+e^- \rightarrow \pi^+\pi^-\gamma$ agree in the total cross section and in every charge-even distribution, but disagree in the pion polar angles: the forward–backward asymmetry of the π^+ is $A(\theta^+) = 0.2375$ in CEEX (and 0.2372 in BabaYaga) against 0.2787 in PHOKHARA. We show that the difference is entirely due to two errors in the one-photon ISR×FSR interference at NLO in PHOKHARA (routine `ampL0`, pion branch of the full-NLO mode): a misplaced parenthesis that adds three helicity terms of the tree interference a second time, and a block that keeps the tree interference together with an analytic ISR soft function that the full-NLO mode replaces by the eikonal elsewhere. The second error makes the PHOKHARA asymmetry depend on the unphysical soft-photon cutoff. A point-by-point comparison of the one-photon matrix elements shows that the two errors account for the whole charge-odd difference to 2.5×10^{-5} of the tree. With both corrected, PHOKHARA gives $A(\theta^+) = 0.23724(10)$ averaged over the cutoff and 0.23725(3) with 5×10^9 events, independent of the cutoff; CEEX (0.23725(3)) and BabaYaga (0.23724) agree with it to 3×10^{-5} .

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*Analysis carried out with Claude Code (Anthropic). Repository: https://github.com/jpaltrinieri/CEEX_PHOKHARA_ANALYSIS

1 Summary of findings

1. **The disagreement is reproduced** with a fresh PHOKHARA build and fresh CEEX runs (Section 4): totals agree to 10^{-4} , charge-even shapes agree, and only $\theta^\pm$ and θ_{avg} differ ($\chi^2/\text{ndf} \approx 10^3$). At LO the two codes agree exactly.
2. **The PHOKHARA asymmetry depends on the soft cutoff w** : $A(\theta^+)$ changes by $+0.0156$ per decade of w while σ is constant (Section 6.1).
3. **The whole effect sits in the charge-odd part of the one-photon contribution** (Born + virtual + soft); the hard two-photon emission agrees between the codes (Section 6.3).
4. **Two errors in PHOKHARA explain it point by point** (Sections 6.4 and 7):
 - **H1** – `phokhara_10.0.f:9969-9982`: the factor $(v_f + v_s)$ multiplies only the last of four helicity terms of the tree interference;
 - **H2** – block B5, the ISR one-loop current times the FSR tree (`phokhara_10.0.f:9998-10007`; `ISRcurrent_no_mass_terms.f:134,210`): an extra $\text{Re}[F^*I](1 + S_{\text{ISR}}(w))$.

Subtracting both leaves a per-point residual of 2.5×10^{-5} of the tree.

5. **With both corrected**, $A(\theta^+)$ no longer depends on w and agrees with CEEX and BabaYaga, in total and separately in the one- and two-photon parts (Section 8). Totals and charge-even distributions are unchanged.

2 Setup

Scenario. KLOE-LA (Strong2020 KLOE-I), $\sqrt{s} = 1.02$ GeV: $50^\circ \leq \theta_\pm \leq 130^\circ$; $|p_\pm^z| > 90$ MeV or $p_\pm^\perp > 160$ MeV; at least one photon with $50^\circ \leq \theta_\gamma \leq 130^\circ$ and $E_\gamma > 20$ MeV; $0.1 \leq M_{\pi\pi}^2 \leq 0.85$ GeV². Angles are measured from the electron beam in both codes. Histograms: the 18 KLOE-I observables with 600 uniform bins. The forward–backward asymmetry is $A = [\sigma(\theta < 90^\circ) - \sigma(\theta > 90^\circ)]/\sigma$.

Codes.

- **CEEX-main** (commit e05d007): `pipig -scenario=KLOE-LA -NLO`, one-loop amplitudes from GoSam (`eepipig_noVP_1L_onshell`, `eepipigg`), YFS soft subtraction with cutoff $E_{\text{min}} = 10^{-5}$ GeV (10^{-4} GeV only in the cutoff checks); 5 MPI ranks per task.
- **PHOKHARA 10.0** (`PHOKARA_10.0 @ 133d68b`) with the KLOE-I cut file, card `nlo=nlo2=1`, `IFSNLO=1`, `ISR+INT+FSR`, `ivac=0`, `FF_pion=-1` ($F_\pi = 1$), no GVMD, no f_0 , soft cutoff $w = 10^{-5}$ (soft region $E_\gamma < w\sqrt{s}$). The production binary `phokhara_nostop` only disables the abort on “Max(i) too small” (Section 10); only weighted events are used.
- **BabaYaga** and the earlier PHOKHARA reference set (5 Aug) are taken from the Strong2020 comparison files.

Pipeline. SLURM arrays per run name; CEEX rank outputs are screened with `check_total.sh` and outliers are moved aside (`quarantine.sh`) before merging; `plot_compare.py` produces the distribution and ratio plots. Everything is in the repository (Appendix A).

3 How PHOKHARA computes $\pi^+\pi^-\gamma$ at NLO

This section describes the pion branch of PHOKHARA 10.0 in the configuration of our card (`pion=1`, `nlo=nlo2=1`, `fsr=2`, `fsrnlo=1`, $F_\pi = 1$, no vacuum polarisation): first the overall structure, then the routines, the UV and IR treatment, and finally the two blocks where the errors are (Section 3.5). Line numbers refer to the unpatched `phokhara_10.0.f` unless another file is named. Statements marked (*inferred*) are our reading of the code; the code itself does not document them.

3.1 Overall structure

Amplitudes. At tree level the one-photon amplitude is $M = I + F$:

- I (ISR): the photon is emitted from the $e^\pm$ line, and $\gamma^*(q^2) \rightarrow \pi\pi$ with the current $J^\mu = (q_+ - q_-)^\mu F_\pi(q^2)$;
- F (FSR): $\gamma^*(s) \rightarrow \pi\pi\gamma$ in scalar QED, i.e. emission from π^+ , from π^- and the seagull (point-like pions).

$|I|^2$ and $|F|^2$ are charge-even; the interference $2\text{Re}[I^*F]$ is charge-odd (odd under $\pi^+ \leftrightarrow \pi^-$) and is the only source of the pion forward–backward asymmetry. At NLO every one-loop and soft term is either charge-even or charge-odd; Table 1 lists them all.

Event classes. Two parameters split the photon phase space:

- w , the soft-photon cutoff: a photon with $E_\gamma < w\sqrt{s}$ (CM frame) is soft, is never generated, and is integrated analytically;
- $\mathbf{gmin}$, the minimum energy of the tagged photon.

Each event is a one-photon or a two-photon event; one random number picks the class with probability proportional to its maximum weight from the scan pass.

- **One-photon events** (one hard photon above $\mathbf{gmin}$) carry in a single weight the tree $|M|^2$, all one-loop virtual corrections, and the integrated soft emission.
- **Two-photon events** carry the exact tree $|M|^2$ for $e^+e^- \rightarrow \pi^+\pi^-\gamma\gamma$: one photon is tagged ($E \geq \mathbf{gmin}$, inside the angular cuts), the other only has to be above $w\sqrt{s}$.

The two classes add up to the full $\mathcal{O}(\alpha)$ correction, and the w dependence must cancel in the sum. This is phase-space slicing; there are no local subtraction terms.

Precision. The one-photon kinematics are generated in quad precision (`momenta_quad.f`). Only `virt_pions` uses the quad momenta; all other pieces use rounded double copies.

3.2 The one-photon weight

Notation. Write the one-photon amplitude to $\mathcal{O}(e^5)$ as

$$M = \underbrace{I + F}_{\text{tree}} + \underbrace{\delta I_e + \delta I_\pi + \delta F_e + \delta F_\pi + \delta B_e + \delta B_\pi}_{\text{one loop}},$$

where the one-loop amplitudes are

- δI_e : ISR with the loop on the electron line (vertex and self-energy corrections around the emitted photon);
- δI_π : ISR with the $\gamma^*\pi\pi$ vertex correction at q^2 ;
- δF_e : FSR with the $e^+e^-\gamma^*$ vertex correction at s ;
- δF_π : FSR with the loop on the pion line (self-energies, vertices, seagull);
- $\delta B_e, \delta B_\pi$: two-photon exchange between the electron and pion lines (boxes and pentagons), with the real photon on the electron or on the pion line.

Figure 1 shows one example diagram for each class.

The two vertex corrections factorise: $\delta F_e = \frac{1}{2}v_s F$ and $\delta I_\pi = \frac{1}{2}v_f I$, with v_s (`ver_s`) and v_f (`ver_f=vertL0`) real $|M|^2$ -level factors.

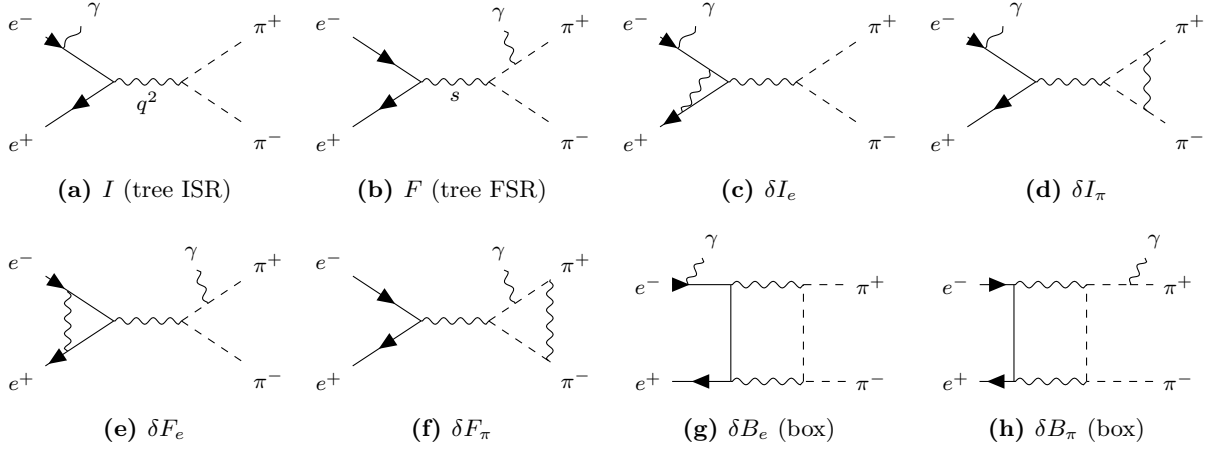


Figure 1: One example diagram per amplitude class of Section 3.2, with the real photon γ ($E_\gamma > w\sqrt{s}$). Top: the tree amplitudes I and F and the ISR corrections; bottom: the FSR corrections and the two-photon exchange. Each class also contains the diagrams with the photon on the other leg, the self-energies, the crossings ($\pi^+ \leftrightarrow \pi^-$, for δB the crossed photon lines) and, for F and δF_π , the seagull $\gamma\gamma\pi\pi$ vertex. The δB examples are boxes (real photon on an external leg); with the real photon on an internal line of the loop they become the pentagons.

From the amplitude expansion to the routines. Squaring M and keeping $\mathcal{O}(\alpha)$ relative to the tree, every term is an interference $2\text{Re}[X^*Y]$ of a tree amplitude $X \in \{I, F\}$ with a tree or one-loop amplitude Y . PHOKHARA computes them in four routines:

$$\begin{aligned}
 |M|^2 = & \underbrace{|I|^2 + 2\text{Re}[I^*\delta I_e]}_{\text{amplit}} \\
 & + \underbrace{|F|^2 + 2\text{Re}[F^*\delta F_e]}_{\text{B1}} + \underbrace{2\text{Re}[I^*\delta I_\pi]}_{\text{B2}} + \underbrace{2\text{Re}[I^*F]}_{\text{B3}} + \underbrace{2\text{Re}[I^*\delta F_e + F^*\delta I_\pi]}_{\text{B4}} + \underbrace{2\text{Re}[F^*\delta I_e]}_{\text{B5}} \\
 & \underbrace{\hspace{10em}}_{\text{helicityampL0}} \\
 & + \underbrace{2\text{Re}[(I+F)^*(\delta F_\pi + \delta B_e + \delta B_\pi)]}_{\text{virt_pions}} + \underbrace{|I+F|^2 \mathcal{S}(w)}_{\text{NLOpions}}, \tag{1}
 \end{aligned}$$

where $\mathcal{S}(w)$ is the eikonal factor for soft emission below $w\sqrt{s}$. The same assignment as a grid, one cell per $2\text{Re}[X^*Y]$ (the F^*I cell is the same term as I^*F):

	I	F	δI_e	δI_π	δF_e	δF_π	$\delta B_e, \delta B_\pi$
I^*	amplit	B3	amplit	B2	B4	virt_pions	virt_pions
F^*	(B3)	B1	B5	B4	B1	virt_pions	virt_pions

The pattern follows from what each routine can do:

- **amplit** is a tensor contraction $L^{\mu\nu}H_{\mu\nu}$, which only exists for $|\text{ISR}|^2$: it gives the I^* entries with I and δI_e , and nothing else.
- The vertex corrections factorise, $\delta F_e = \frac{1}{2}v_s F$ and $\delta I_\pi = \frac{1}{2}v_f I$, so every cell with them is a tree helicity sum times v_s or v_f . Together with the tree $|F|^2$ and $2\text{Re}[I^*F]$ they are all in **helicityampL0** (B1–B4).
- δI_e appears twice. Against I^* it is in **amplit** (analytic, m_e kept); against F^* it needs the amplitude, which **helicityampL0** takes from **NLO_ISR_no_mass_terms** (massless electrons) in B5.
- The non-factorising loops δF_π , δB_e , δB_π are genuine one-loop amplitudes, computed in **virt_pions** against the full tree $I + F$.

Table 1 lists the same terms with their charge factors. Counting charges ($I \propto Q_e^2 Q_\pi$, $F \propto Q_e Q_\pi^2$, each loop adds two charges), a term is charge-odd when it has an odd power of Q_π . The charge-odd cells are I^*F , $I^*\delta F_e$, $F^*\delta I_\pi$, $F^*\delta I_e$, $I^*\delta F_\pi$, $I^*\delta B_e$ and $F^*\delta B_\pi$, plus the odd part of the soft term.

Table 1: Every contribution to the one-photon weight and where PHOKHARA computes it (full-NLO pion mode). B1–B5 are the blocks of `ampL0` (Table 2). $S_{\text{ISR}}(w)$ is the analytic ISR soft function of Section 3.4. The hard two-photon emission, the other half of the real correction, is computed separately in `gen_2ph`.

Contribution	Charges	C	Computed in	Remark
<i>Tree</i>				
$ I ^2$	$Q_e^4 Q_\pi^2$	+	<code>amplit</code>	tensor $L^{\mu\nu} H_{\mu\nu}$
$ F ^2$	$Q_e^2 Q_\pi^4$	+	<code>ampL0</code> B1	
$2\text{Re}[I^*F]$	$Q_e^3 Q_\pi^3$	–	<code>ampL0</code> B3	only with <code>fsr=2</code>
<i>One loop</i>				
$2\text{Re}[I^*\delta I_e]$	$Q_e^6 Q_\pi^2$	+	<code>amplit</code>	Rodrigo, m_e kept; $-S_{\text{ISR}} I ^2$
$2\text{Re}[I^*\delta I_\pi] = v_f I ^2$	$Q_e^4 Q_\pi^4$	+	<code>ampL0</code> B2	
$2\text{Re}[F^*\delta F_e] = v_s F ^2$	$Q_e^4 Q_\pi^4$	+	<code>ampL0</code> B1	
$2\text{Re}[F^*\delta F_\pi]$	$Q_e^2 Q_\pi^6$	+	<code>virt_pions</code>	<code>virtual_FSRHelSum</code>
$2\text{Re}[I^*\delta F_e + F^*\delta I_\pi]$ $= \text{Re}[I^*F](v_s + v_f)$	$Q_e^5 Q_\pi^3, Q_e^3 Q_\pi^5$	–	<code>ampL0</code> B4	H1
$2\text{Re}[F^*\delta I_e]$	$Q_e^5 Q_\pi^3$	–	<code>ampL0</code> B5	massless e ; H2
$2\text{Re}[I^*\delta F_\pi]$	$Q_e^3 Q_\pi^5$	–	<code>virt_pions</code>	<code>virtual_FSRHelSum</code>
$2\text{Re}[I^*\delta B_e]$	$Q_e^5 Q_\pi^3$	–	<code>virt_pions</code>	<code>Pentabox_full_isr</code> , <code>cross_ISR</code>
$2\text{Re}[F^*\delta B_e]$	$Q_e^4 Q_\pi^4$	+	<code>virt_pions</code>	same routines
$2\text{Re}[I^*\delta B_\pi]$	$Q_e^4 Q_\pi^4$	+	<code>virt_pions</code>	<code>Pentabox_PIFSR</code> , <code>cross_FSR</code>
$2\text{Re}[F^*\delta B_\pi]$	$Q_e^3 Q_\pi^5$	–	<code>virt_pions</code>	same routines
<i>Soft emission, $E_\gamma < w\sqrt{s}$</i>				
$ M_{\text{tree}} ^2 \times (ee, \pi\pi \text{ eikonal})$	$\times Q_e^2, \times Q_\pi^2$	$\pm$	<code>NLOpions</code>	odd part from $2\text{Re}[I^*F]$
$ M_{\text{tree}} ^2 \times (e\pi \text{ eikonal})$	$\times Q_e Q_\pi$	$\mp$	<code>NLOpions</code>	odd part from $ I ^2 + F ^2$

amplit: first line of Equation (1) (lines 2069, 6757ff). `amplit` = $L^{\mu\nu} H_{\mu\nu}$ is a contraction of a leptonic and a hadronic tensor, not a helicity sum:

- $H^{\mu\nu} \propto |F_\pi(q^2)|^2 (q_+ - q_-)^\mu (q_+ - q_-)^\nu$ (`HadronicTensorISR`): the $\gamma^* \rightarrow \pi\pi$ vertex, squared;
- $L^{\mu\nu}$ (`LeptonicTensor1ISR`): the electron line with the emitted photon, squared and summed over the photon polarisation. Its coefficients come from `Bornvirtualsoft` (G. Rodrigo et al., EPJC 22 (2001) 81; J.H. Kühn, G. Rodrigo, EPJC 25 (2002) 215): LO plus the $\mathcal{O}(\alpha)$ ISR virtual+soft correction with m_e kept, a renormalised analytic result.

With `nlo2=1` the analytic soft function is then removed (lines 6838–6856), so

$$\text{amplit} = |I|^2 + 2\text{Re}[I^*\delta I_e]_{\text{V+S}} - S_{\text{ISR}}(w) |I|^2 = |I|^2 + 2\text{Re}[I^*\delta I_e]_{\text{V}},$$

where the subscript V means the virtual correction with the IR pole dropped and its soft counterpart supplied by `NLOpions`. `amplit` contains nothing else: no FSR, no interference, and not the pion-vertex correction (that is B2).

helicityampL0 $\rightarrow$ **ampL0:** second line of Equation (1) (lines 9687–10021). One block per underbrace:

$$\text{helicityampL0} = \frac{(4\pi\alpha)^3}{4} \text{ampL0} \times (\text{phase space}), \quad \text{ampL0} = \sum_{\lambda_\gamma=1,2} (B_1 + B_2 + B_3 + B_4 + B_5),$$

the $1/4$ being the $e^\pm$ spin average. The amplitudes are 2×2 Weyl blocks A and B, `plus`(v) = $v^0 - \vec{v} \cdot \vec{\sigma}$ and `minus`(v) = $v^0 + \vec{v} \cdot \vec{\sigma}$:

Amplitude	A block	B block
I (tree ISR, from <code>blocksL0</code>)	<code>ma</code>	<code>mb</code>
F (tree FSR, from <code>ddvec</code>)	<code>ddmi</code>	<code>ddpl</code>
I_{1L} (ISR at one loop, from <code>NLO_ISR_no_mass_terms</code>)	<code>ami</code>	<code>apl</code>

The beam spinors enter through four fixed combinations of the blocks (lines 9939–9942), with $e_b = E + |\vec{p}|$ of the positron:

$$\begin{aligned} T_1(X) &= m_e(B_{11} - A_{11}), & T_3(X) &= -e_b A_{21} + \frac{m_e^2}{e_b} B_{21}, \\ T_2(X) &= m_e(B_{22} - A_{22}), & T_4(X) &= e_b B_{12} - \frac{m_e^2}{e_b} A_{12}. \end{aligned}$$

(Inferred) T_1, T_2 are the m_e -suppressed helicity-flip amplitudes and T_3, T_4 the two leading helicity-conserving ones. With $\langle X, Y \rangle \equiv \sum_{k=1}^4 T_k(X)^* T_k(Y)$ and $|X|^2 \equiv \langle X, X \rangle$, the five blocks are listed in Table 2.

Table 2: The blocks of `ampL0` (pion branch), per photon polarisation. Vacuum-polarisation factors `cvac` are 1 in our setup and omitted.

Block	Code computes	Physics content	Condition	Lines
B1	$ F ^2(1 + v_s)$	$ F ^2 + 2\text{Re}[F^* \delta F_e]$	always ^a	9938–9943
B2	$ I ^2 v_f$	$2\text{Re}[I^* \delta I_\pi]$	<code>fsrcno=1</code>	9945–9951
B3	$2\text{Re}\langle I, F \rangle$	$2\text{Re}[I^* F]$	<code>fsrc=2</code>	9954–9966
B4	$\text{Re} \sum_{k \leq 3} T_k(I)^* T_k(F)$ $+ \text{Re}[T_4(I)^* T_4(F)](v_f + v_s)$	intended $\text{Re}\langle I, F \rangle(v_f + v_s)$	<code>nlo2=1, pion=1</code>	9969–9982 H1
B5	$\text{Re}\langle F, I_{1L} \rangle$, with $I_{1L} = I(1 + S_{\text{ISR}}) + 2\delta I_e$	intended $2\text{Re}[F^* \delta I_e]$	<code>nlo2=1, pion=1</code>	9984–10007 H2

^a $v_s = 0$ if `fsrcno=0`.

As coded, B4 and B5 differ from their underbraces in Equation (1) by two extra terms, both proportional to the tree interference:

$$\Delta B_4 = \text{Re} \sum_{k \leq 3} T_k(I)^* T_k(F) (1 - (v_f + v_s)), \quad \Delta B_5 = \text{Re}\langle I, F \rangle (1 + S_{\text{ISR}}(w)).$$

The decomposition $I_{1L} = I(1 + S_{\text{ISR}}) + 2\delta I_e$ is read from `V1(1)` (Section 7.2) and confirmed numerically by the per-point test (Section 6.4).

NLOptions: last term of Equation (1) (`hard_soft_part/one_photon.f`).

$$|M_{\text{tree}}|^2 \times 2 [S_{jj}^{ee} - S_{12}^{ee} + S_{jj}^{\pi\pi} - S_{12}^{\pi\pi} + (-S_{11}^{e\pi} - S_{22}^{e\pi} + S_{12}^{e\pi} + S_{21}^{e\pi})],$$

with the full tree $|M_{\text{tree}}|^2 = |I|^2 + |F|^2 + 2\text{Re}\langle I, F \rangle$ recomputed in helicity form, and the eikonal integrals of `soft_integrals.f` ('t Hooft–Veltman, eq. (7.3)) evaluated on the event momenta: `pjpj`, `p1p2` (ee), `qjqj_sub`, `q1q2_sub` ($\pi\pi$) and the four $e\pi$ interference terms.

virt_pions: third line of Equation (1) (`virtualpions.f`, quad precision, $\mu^2 = s$). Each loop amplitude is contracted with a helicity vector of the full tree $I + F$ (the `hel()` arrays carry both the $F_\pi(s)$ and the $F_\pi(q^2)$ parts), so each routine gives both the charge-even and the charge-odd rows of Table 1. The diagram content is inferred from the propagator masses:

- `Pentabox_full_isrHelSum`, `cross_ISRHelSum`: δB_e , masses $(m_e, m_e, 0, m_\pi, 0)$, and its $\pi^+ \leftrightarrow \pi^-$ crossing;
- `Pentabox_PIFSRHelSum`, `cross_FSRHelSum`: δB_π , masses $(m_e, 0, m_\pi, m_\pi, 0)$, and its crossing;
- `virtual_FSRHelSum` and `...cr (test_vFSR.f)`: δF_π , only massless and m_π propagators, with QCDLoop integrals.

3.3 UV renormalisation

- **Analytic pieces.** Bornvirtualsoft, v_s and v_f (`vertL0`) are closed-form, on-shell renormalised results: electron vertex at s and point-like pion form factor at q^2 , normalised to the charge e at $q^2 = 0$. No counterterm appears in the code; with vacuum polarisation off, α does not run.
- **Pion-line loops** (`test_vFSR.f:740–761`). This is the only place with explicit counterterms: on-shell scalar-QED counterterms built from B_0 , C_0 and the derivative $B'_0(m_\pi^2)$ (wave-function renormalisation), multiplied by tree helicity structures (`diag1`, `diag2`) and added to the result (`counter1`, `counter3`, `counter8`, `counter95`). The crossed version has no `diag1` analogue.
- **Boxes and pentagons.** The two-photon-exchange set is UV-finite and has no counterterms. Tensor integrals are reduced with Passarino–Veltman in quad precision; scalar integrals come from QCDLoop.
- **Regularisation.** Dimensional regularisation with $\mu^2 = s$; the $1/\epsilon$ poles are dropped and the finite parts kept. The IR poles cancel against the soft integrals, written in the same convention with $\lambda = \sqrt{s}$. Scale shifts give $\ln(q^2/s)$ terms, e.g. the extra $-\frac{1}{2}\ln(q^2/s)$ in `qjqj_sub` and `q1q2_sub`.

3.4 IR treatment

Slicing. Soft region $E_\gamma < w\sqrt{s}$ (CM frame): eikonal approximation, integrated analytically in `NLOptions`. Hard region: exact two-photon matrix element (`gen_2ph`), with the ISR·ISR (`ma/mb`), ISR·FSR (`mapi`) and FSR·FSR (`Jplus/Jmin`) amplitudes added coherently and squared with the same T_1 – T_4 pattern.

The full-NLO rewrite (`nlo2=1`). In the older NLO mode each analytic vertex result was “virtual + soft”, with its own analytic $\ln(2w)$ soft function (ISR in the leptonic tensor, v_s for the electron vertex, `vertL0` for the pion vertex). That scheme has no e – π soft interference. `nlo2=1` replaces all soft emission by the exact eikonal integrals of `NLOptions`, which include the $e\pi$ terms, so every analytic soft function has to be removed from its “V+S” formula. The code does this in three places:

- leptonic tensor (lines 6838–6856): $L - L_{\text{LO}} S_{\text{ISR}}(w)$, with

$$S_{\text{ISR}}(w) = \frac{\alpha}{\pi} \left[-\ln(4w^2) (1 + L) - \frac{1}{2}L^2 - L - \frac{\pi^2}{3} \right], \quad L = \ln \frac{m_e^2}{s};$$

- v_s (lines 2754–2761): $v_s - S_{\text{ISR}}(w)$; the $\ln(2w)$ terms cancel exactly;
- `vertL0` (line 9795): virtual part minus the soft function of Nucl. Phys. B57 (1973) 381, eq. (29), “without terms proportional to $\text{Log}(2w)$ ”.

What must hold. After the rewrite every one-loop piece should be free of analytic w terms. All w dependence then comes from `NLOptions` and cancels against the two-photon events. A leftover analytic $\ln w$ term does not cancel, and the result depends on w .

3.5 Where the errors are

Both errors are in `ampL0`, in the two blocks of Table 2 that correct the tree interference, and both add a multiple of the tree interference (the ΔB_4 , ΔB_5 of Section 3.2):

- **B4 (H1).** The vertex factor ($v_f + v_s$) is placed inside `dreal()` after the last helicity term, so it multiplies only T_4 . T_1 – T_3 of the tree interference are added a second time with coefficient 1. T_3 is a leading helicity amplitude, so this is an $\mathcal{O}(1)$ extra charge-odd term, not an $\mathcal{O}(\alpha)$ one.

- **B5 (H2).** `NLO_ISR_no_mass_terms` returns the ISR amplitude *including* the tree, not the loop alone: its leading coefficient `V1(1)` contains the tree “1” and the analytic soft function $S_{\text{ISR}}(w)$. The “1” repeats half of B3, and $S_{\text{ISR}}(w)$ is the one soft function the full-NLO rewrite never removed, which breaks the condition of Section 3.4.

Both extra terms are proportional to $\text{Re}\langle I, F \rangle$, which is charge-odd. They integrate to zero over the charge-symmetric KLOE-LA cuts, so σ and all charge-even distributions are unaffected; only the pion polar angles move. Section 7 gives the code and the fixes.

Oddities without effect here. `soft_f` in `vertL0` and `ver_f` in `ampNLO` are computed but not used. The virtual-FSR term in `virt_pions` has an explicit prefactor four times that of the pentagons ($32\pi^2\alpha^4$ vs $8\pi^2\alpha^4$); we did not check whether this is compensated inside the routine, but the per-point test (Section 6.4) agrees with CEEX to 2.5×10^{-5} of the tree. Some “(-1)” comments do not match the code.

4 The discrepancy

Table 3: Before the fix. PHOKHARA (unpatched) 300 seeds $\times$ 500k events; CEEX more1M, 148 tasks $\times$ 5 ranks $\times$ 1M events at $E_{\text{min}} = 10^{-5}$ GeV (732 rank outputs after discarding 8 whose total diverged).

	σ [nb]	$A(\theta^+)$
CEEX, $E_{\text{min}} = 10^{-5}$ GeV	0.268389(9)	0.23725(3)
BabaYaga	0.268389(4)	0.23724(1)
PHOKHARA, this work	0.268353(46)	0.27872(16)
PHOKHARA, earlier reference	0.268393(23)	0.27861(8)

The new PHOKHARA run reproduces the earlier reference. All generators pass the charge-conjugation mirror test $\theta^+(\theta) = \theta^-(180^\circ - \theta)$, so this is not a mislabelled charge. Figure 2 shows the θ^+ distribution and Figure 3 the ratio of every observable to PHOKHARA: only the three pion angles are off, by a smooth tilt from 0.95 to 1.28.

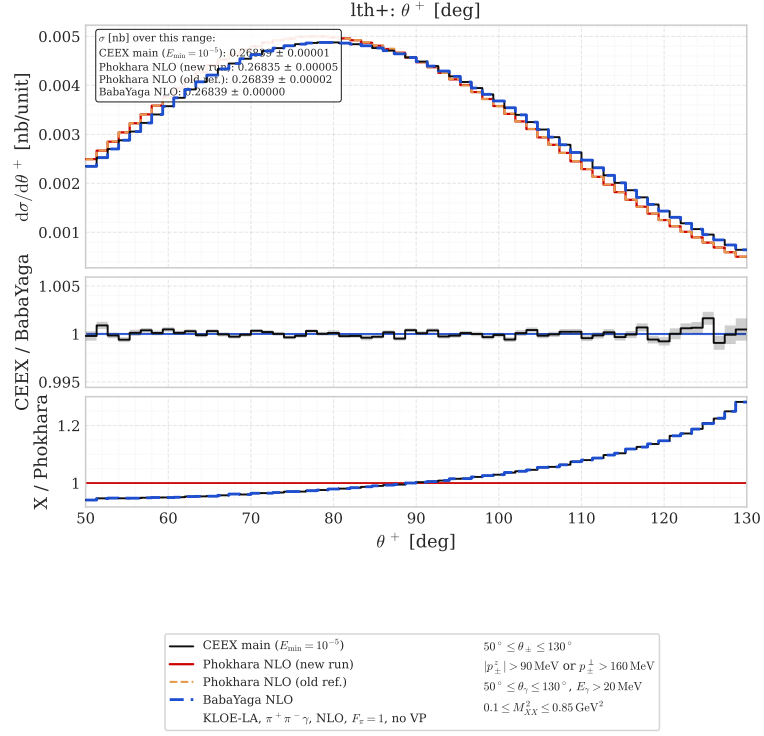


Figure 2: Before the fix: unpatched PHOKHARA (300×500k) against CEEEX more1M (148×5×1M): $d\sigma/d\theta^+$, CEEEX/BabaYaga, and CEEEX and BabaYaga over PHOKHARA (60 bins). CEEEX ($E_{\min} = 10^{-5}$ GeV, black) and BabaYaga (blue) agree; PHOKHARA (red) is tilted.

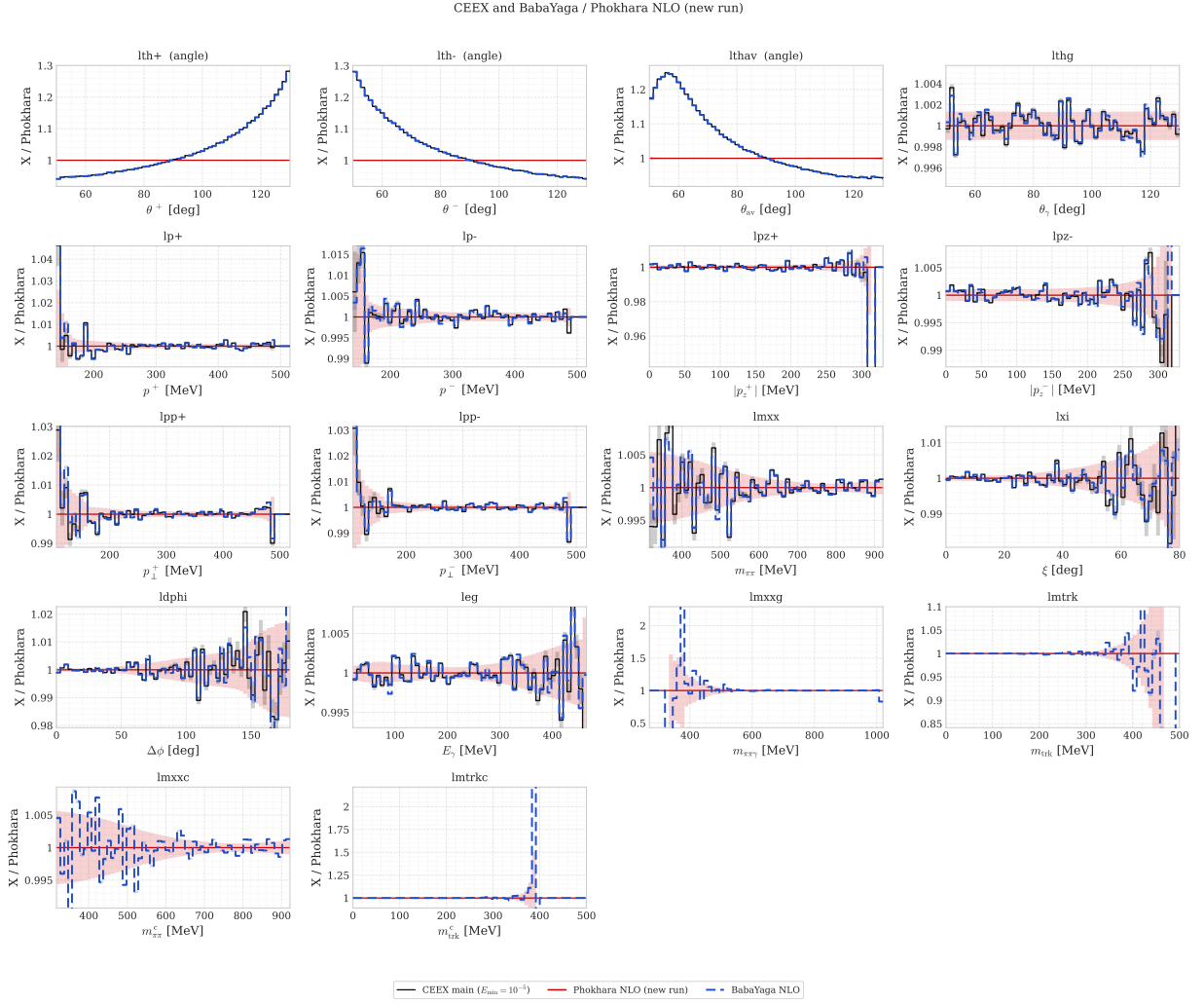


Figure 3: Before the fix: every observable, CEEX and BabaYaga over PHOKHARA. Only θ^+ , θ^- and θ_{avg} are off; the spikes in $M_{\pi\pi\gamma}$ and M_{trk} are empty tail bins.

At LO the codes agree exactly (earlier runs with the Strong2020 form factor: $A_{\text{LO}} = 0.0779$ and $\sigma_{\text{LO}} = 2.6523$ nb in both), so the difference lies in the NLO correction to the charge-odd part of the cross section.

5 Strategy

The difference is charge-odd (C -odd): it flips sign under $\pi^+ \leftrightarrow \pi^-$ and integrates to zero over the symmetric cuts. At $\mathcal{O}(\alpha^4)$ the C -odd part of the cross section consists of the gauge-invariant classes $Q_e^5 Q_\pi^3$ and $Q_e^3 Q_\pi^5$ (the tree interference is $Q_e^3 Q_\pi^3$). Three tools were used to isolate it:

- **Cutoff independence.** The split into one-photon (Born + virtual + soft photons below the cutoff) and two-photon (hard) parts depends on the cutoff; the sum must not.
- **Matched split.** For the same cutoff each part is uniquely defined, so the two codes must agree part by part. PHOKHARA $w = 9.804 \times 10^{-6}$ ($w\sqrt{s} = 10^{-5}$ GeV) matches CEEX $E_{\min} = 10^{-5}$ GeV.
- **Momentum swap.** At a phase-space point, the C -odd part of any piece X is $X_{\text{odd}} = \frac{1}{2}[X(p_+, p_-) - X(p_-, p_+)]$. No charge parameter is needed.

Each histogram-level test used enough statistics to resolve $A(\theta^+)$ to $\sim 10^{-3}$ while checking that σ and $M_{\pi\pi}$ are unchanged. The plan with all tests and hypotheses is in `plan.txt`.

6 Tests

6.1 T1: soft-cutoff scan

Table 4: Unpatched PHOKHARA, 30 seeds $\times$ 100k per value of w .

w	10^{-3}	10^{-4}	10^{-5}	10^{-6}
$\sigma_{1\gamma}$ [nb]	0.1386	0.0955	0.0530	0.0103
$\sigma_{2\gamma}$ [nb]	0.1297	0.1728	0.2149	0.2580
σ [nb]	0.26833(33)	0.26827(35)	0.26792(33)	0.26830(30)
$A(\theta^+)$	0.3106(11)	0.2950(12)	0.2785(12)	0.2637(11)

$A(\theta^+)$ falls linearly in $\ln w$, by 0.0156 per decade, while σ is constant to 0.1%: the charge-even soft logarithms cancel between the one- and two-photon parts, a charge-odd one does not. In CEEX, $E_{\min} = 10^{-4}$ and 10^{-5} GeV give the same $A(\theta^+)$ to 10^{-4} .

6.2 T2: diagnostic variants

Three diagnostic PHOKHARA builds were run at $w = 10^{-4}, 10^{-5}, 10^{-6}$ (Figure 4a):

- **H1 fixed** (the parenthesis of Section 7.1): $A(\theta^+)$ moves by -0.037 and keeps its slope. H1 is a w -independent error.
- **B5 soft log removed** (probe): $A(\theta^+)$ becomes flat in w , but at 0.355. The whole w -dependence comes from this term; removing only the logarithm is not the right correction.
- **Both:** flat, at 0.319.

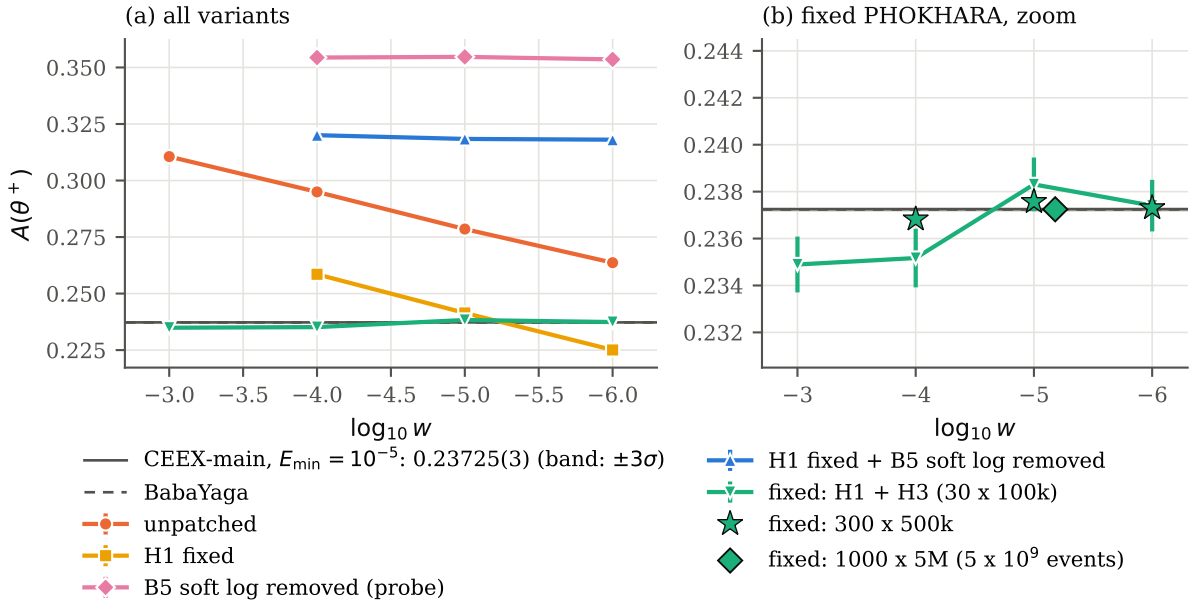


Figure 4: $A(\theta^+)$ against the PHOKHARA soft cutoff for the unpatched code, the T2 diagnostic variants and the fixed code (H1 + H3), with CEEX (grey band: $\pm 3\sigma$) and BabaYaga. The zoom (b) shows the fixed code at $30 \times 100k$, at $300 \times 500k$ (stars) and at $1000 \times 5M$ (diamond, drawn just left of $w = 10^{-5}$).

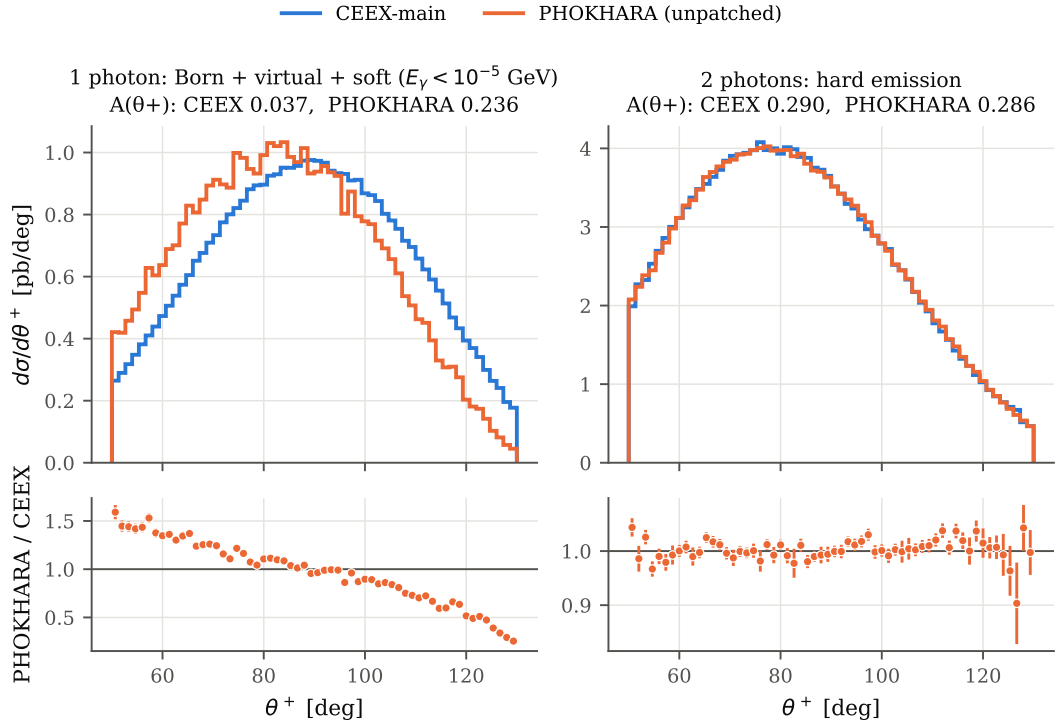
6.3 T3: split by photon multiplicity

CEEX-main was given an opt-in output of the per-multiplicity histograms (CEEX_NPH_HISTOS=1); PHOKHARA writes them with `print_summed=1`.

Table 5: Matched cutoff (10^{-5} GeV). CEEX 10 seeds $\times$ 5 ranks $\times$ 100k; PHOKHARA 30 $\times$ 100k (unpatched).

	σ [nb]		$A(\theta^+)$		angle shapes
	CEEX	PHOKHARA	CEEX	PHOKHARA	χ^2/ndf
1 photon (Born+V+S)	0.05268(7)	0.05239(24)	0.0375(13)	0.2360(44)	79–94
2 photons (hard)	0.21560(37)	0.21588(23)	0.2901(16)	0.2855(10)	1.0–1.4

The one-photon parts have the same σ and the same $M_{\pi\pi}$, θ_γ and E_γ shapes, but charge-odd cross sections of 0.0020 (CEEX) and 0.0124 nb (PHOKHARA). The difference, 0.0104 nb, is the whole total discrepancy (Figure 5). The hard two-photon parts agree.

**Figure 5:** T3: $d\sigma/d\theta^+$ of the one-photon (left) and two-photon (right) contributions at a matched cutoff, with the ratio PHOKHARA/CEEX.

6.4 T4: point-by-point comparison

988 accepted one-photon KLOE-LA points were dumped by a PHOKHARA variant (`phokhara_dump`) together with every piece of the one-photon matrix element, at the event and with the pion momenta swapped. A CEEX-main driver (`t4_ceex`) evaluates the tree, the hard virtual $\frac{\alpha}{4\pi}D$ (GoSam finite part plus mass counterterm), the wave-function counterterm and the YFS factor at the same momenta; the pion mass is set to PHOKHARA's value. Each code is normalised to its own charge-even tree:

- tree C -odd/ C -even ratio: identical to 7×10^{-13} ;
- C -even NLO correction: CEEX -0.77566 , PHOKHARA -0.77570 , per-point difference rms 10^{-5} ;
- C -odd NLO correction: per-point difference D with rms 0.099;
- C -odd soft (eikonal/YFS) part: agrees, rms 7×10^{-4} .

D is exactly proportional to the tree interference at every point. Subtracting the two errors, each computed exactly per point ($v_s = 0.29655$, $v_f = 0.023\text{--}0.036$, $S_{\text{ISR}}(w) = -0.95556$),

$$\text{pred}_{\text{H1}} = B_4 - \frac{1}{2}B_3(v_f + v_s), \quad \text{pred}_{\text{H2}} = \frac{1}{2}B_3(1 + S_{\text{ISR}}(w)),$$

leaves a residual of rms 2.5×10^{-5} with mean 10^{-6} (Figure 6). H1 alone leaves 1.2×10^{-2} , H2 alone 8.8×10^{-2} .

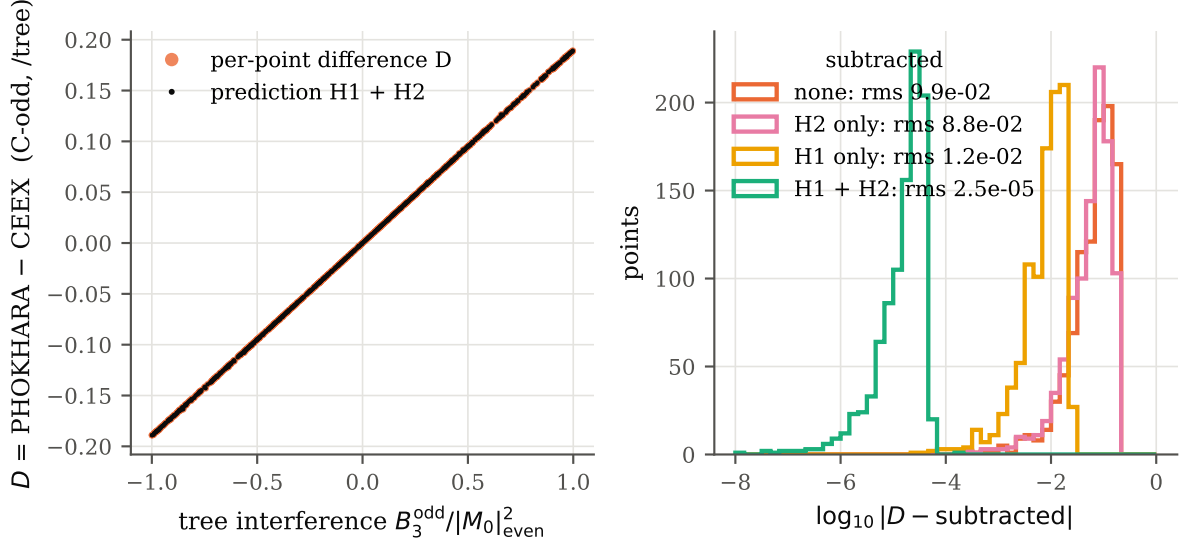


Figure 6: T4, 988 points. Left: the C -odd difference D against the tree interference, with the H1+H2 prediction. Right: $\log_{10}|D - \text{subtracted}|$ for no subtraction, H1, H2 and H1+H2.

7 The two errors in PHOKHARA

The structure of `ampL0` and the blocks B1–B5 are described in Section 3.2 and Table 2. Here I and F are the ISR and FSR tree amplitudes, v_s the electron-vertex correction at s and v_f the pion-vertex correction at q^2 .

7.1 H1: misplaced factor in the $\text{ISR} \times \text{FSR}$ correction

```

if((nlo2.eq.1).and.(pion.eq.1))then
  ampL0 = ampL0
4    + dreal(
5      dconjg(dme*(mb(1,1)-ma(1,1))*cvac_qq)*dme*(ddpl(1,1)-ddmi(1,1))*cvac_s    ! T1
6      + dconjg(dme*(mb(2,2)-ma(2,2))*cvac_qq)*dme*(ddpl(2,2)-ddmi(2,2))*cvac_s    ! T2
7      + dconjg((-ebppb*ma(2,1)+e1_m2/ebppb*mb(2,1))*cvac_qq)
8      *(-ebppb*ddmi(2,1)+e1_m2/ebppb*ddpl(2,1))*cvac_s                          ! T3
9      + dconjg((ebppb*mb(1,2)-e1_m2/ebppb*ma(1,2))*cvac_qq)
1     *(ebppb*ddpl(1,2)-e1_m2/ebppb*ddmi(1,2))*cvac_s                          ! T4
2     *(ver_f+ver_s) )                  ! multiplies T4 only

```

(Continuation lines re-indented for reading; the code is unchanged.) The block is meant to be $\text{Re}[T_1 + T_2 + T_3 + T_4](v_f + v_s)$, the amplitude-level corrections on each side of $2\text{Re}[I^*F]$. As written it is $\text{Re}[T_1 + T_2 + T_3] + \text{Re}[T_4](v_f + v_s)$, so $\text{Re}[T_1 + T_2 + T_3]$ (about a third of the tree interference) is added with coefficient 1. The muon version of the same block (`phokhara_10.0.f:10384-10394`) is correct: `2*dreal(sum)*(ver_s+ver_f)/2`.

Fix (variants/h1.patch): `*cvac_s *(ver_f+ver_s) ) → *cvac_s )*(ver_f+ver_s)`.

7.2 H2: tree and analytic ISR soft function in the ISR one-loop block

Block B5 contracts the FSR tree with the ISR current at one loop:

```
ampL0=ampL0+dreal( dconjg(... ddpl, ddmi ...)*( ... apl, ami ...)) ! B5
```

The current (`hard_soft_part/ISRcurrent_no_mass_terms.f`) has the coefficient

```
V1(1)=2.d0*log(4.d0*w**2)*(log(m2)+1.d0) -4.d0*log(qq2)+3.d0*log(m2) + ... ! 1.134
V1(1)=(1.d0-2.d0*alpha/4.d0/pi*V1(1))/Sp ! 1.210
```

It therefore contains the tree (the “1”) and the analytic ISR soft function, and B5 carries an extra $\text{Re}[F^*I](1 + S_{\text{ISR}}(w))$ with

$$S_{\text{ISR}}(w) = \frac{\alpha}{\pi} \left[-\ln(4w^2) (1 + L) - \frac{1}{2}L^2 - L - \frac{\pi^2}{3} \right], \quad L = \ln \frac{m_e^2}{s}.$$

- The tree interference is already added by the `fstr=2` block (B3).
- In the full-NLO mode the analytic ISR soft function is replaced by the eikonal in `NLOptions`; it is subtracted from v_s , v_f and the leptonic tensor, but not here.
- At $w \simeq 10^{-5}$ the two parts nearly cancel ($1 + S_{\text{ISR}} = 0.044$), but S_{ISR} changes by 0.152 per decade of w . This gives the T1 slope: $\frac{1}{2} \cdot 0.152 \cdot A_{\text{LO}} \approx 0.019$ per decade, against 0.0156 measured.

Fix (`variants/h3.patch`): subtract $\text{Re}[\sum T](1 + S_{\text{ISR}})$ after B5, with S_{ISR} computed as in the v_s subtraction.

8 Result with the fix

`phokhara_fix` is the production binary with `h1.patch` and `h3.patch`.

Table 6: Fixed PHOKHARA against the soft cutoff.

w	10^{-3}	10^{-4}	10^{-5}	10^{-6}
<i>30 seeds $\times$ 100k</i>				
σ [nb]	0.26799(33)	0.26892(35)	0.26854(32)	0.26784(31)
$A(\theta^+)$	0.2349(12)	0.2352(13)	0.2383(12)	0.2374(11)
<i>300 seeds $\times$ 500k</i>				
σ [nb]	–	0.268345(48)	0.268351(46)	0.268354(44)
$A(\theta^+)$	–	0.23682(17)	0.23757(17)	0.23730(16)
<i>1000 seeds $\times$ 5M (5×10^9 events)</i>				
σ [nb]	–	–	0.268398(8)	–
$A(\theta^+)$	–	–	0.23725(3)	–

At high statistics the slope in w is -0.00023 ± 0.00012 per decade (unpatched: $+0.0156$), and the w -averaged asymmetries are

	$A(\theta^+)$	$-A(\theta^-)$	$-A(\theta_{\text{avg}})$
PHOKHARA, fixed (300 $\times$ 500k, w -average)	0.23724(10)	0.23722(10)	0.26651(9)
PHOKHARA, fixed (1000 $\times$ 5M, $w = 10^{-5}$)	0.23725(3)	0.23728(3)	0.26651(3)
CEEX, $E_{\text{min}} = 10^{-5}$ GeV (more1M, 148 $\times$ 5 $\times$ 1M)	0.23725(3)	0.23726(3)	0.26648(3)
BabaYaga	0.23724(1)	0.23727(1)	0.26650(1)

At $w = 10^{-5}$ and 10^{-6} the fixed PHOKHARA agrees with CEEX and BabaYaga within 0.3 – 1.5σ . The $w = 10^{-4}$ sample lies 0.0007 lower. To check it, the multiplicity split of Section 6.3 was repeated with the fixed code at a matched cutoff of 10^{-4} GeV (PHOKHARA $w = 9.804 \times 10^{-5}$, 100 seeds $\times$ 500k; CEEX $E_{\text{min}} = 10^{-4}$ GeV):

	CEEX	σ [nb] PHOKHARA (fixed)	CEEX	$A(\theta^+)$ PHOKHARA (fixed)
1 photon (Born+V+S)	0.09533(10)	0.09539(7)	0.1548(11)	0.1558(7)
2 photons (hard)	0.17290(23)	0.17301(5)	0.2826(13)	0.2829(2)
sum	0.26823(25)	0.26840(8)	0.2372(9)	0.2377(3)

Both parts agree, with angle-shape $\chi^2/\text{ndf} = 0.7\text{--}1.4$, and this sample gives $A(\theta^+) = 0.2377(3)$, consistent with $w = 10^{-5}$ and 10^{-6} . A 2% change of the cutoff cannot move $A(\theta^+)$ by 0.0009, so the low $w = 10^{-4}$ value is a fluctuation of that sample (2.6σ from this one). T4 at $w = 10^{-4}$ also leaves no systematic C -odd residual after H1+H2 (mean -7×10^{-6}).

Highest statistics. A final run of 1000 seeds $\times$ 5M events (5×10^9 events passing the cuts, $w = 10^{-5}$) gives $\sigma = 0.268398(8)$ nb and $A(\theta^+) = 0.23725(3)$, $-A(\theta^-) = 0.23728(3)$, $-A(\theta_{\text{avg}}) = 0.26651(3)$. To match this precision on the CEEX side, a further CEEX run (more1M: 5 MPI ranks $\times$ 1M events per task at $E_{\text{min}} = 10^{-5}$ GeV, seeds 101–300, independent of prod500k) was started; the 148 tasks finished at the time of writing (732 rank outputs; 8 whose total diverged were discarded by `check_total.sh`) give $\sigma = 0.268389(9)$ nb and

$$A(\theta^+) = 0.23725(3), \quad -A(\theta^-) = 0.23726(3), \quad -A(\theta_{\text{avg}}) = 0.26648(3),$$

in agreement with the fixed PHOKHARA and with BabaYaga to 3×10^{-5} in all three asymmetries. Figures 7 and 8 repeat Figures 2 and 3 with these two runs: the pion-angle shape χ^2/ndf of CEEX against the fixed PHOKHARA drops from ~ 1900 to 0.8–1.4, and against BabaYaga it is 0.9–1.2.

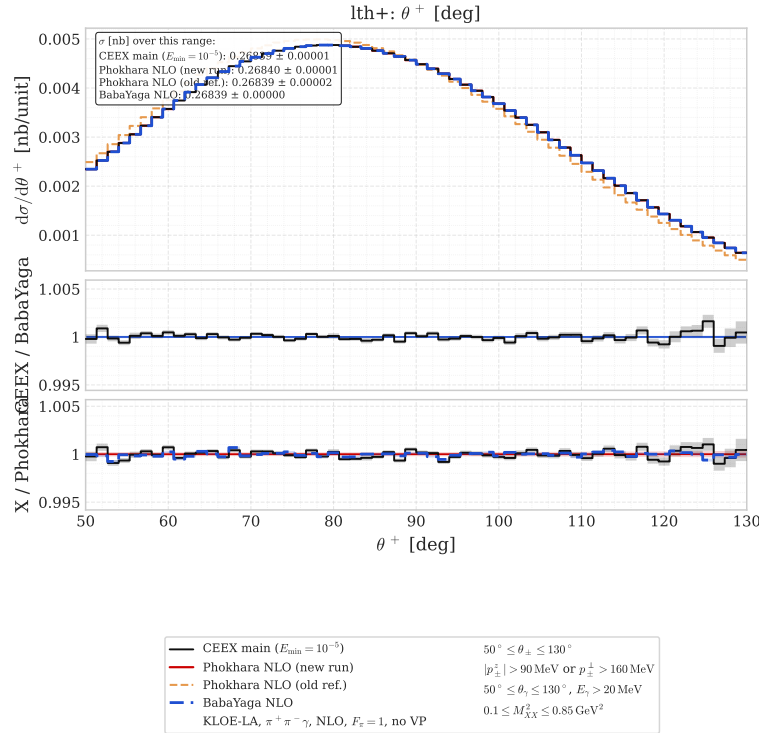


Figure 7: After the fix: as Figure 2, with the fixed PHOKHARA (1000 $\times$ 5M, $w = 10^{-5}$) as the denominator and CEEX more1M (148 $\times$ 5 $\times$ 1M, $E_{\text{min}} = 10^{-5}$ GeV). The tilt is gone.

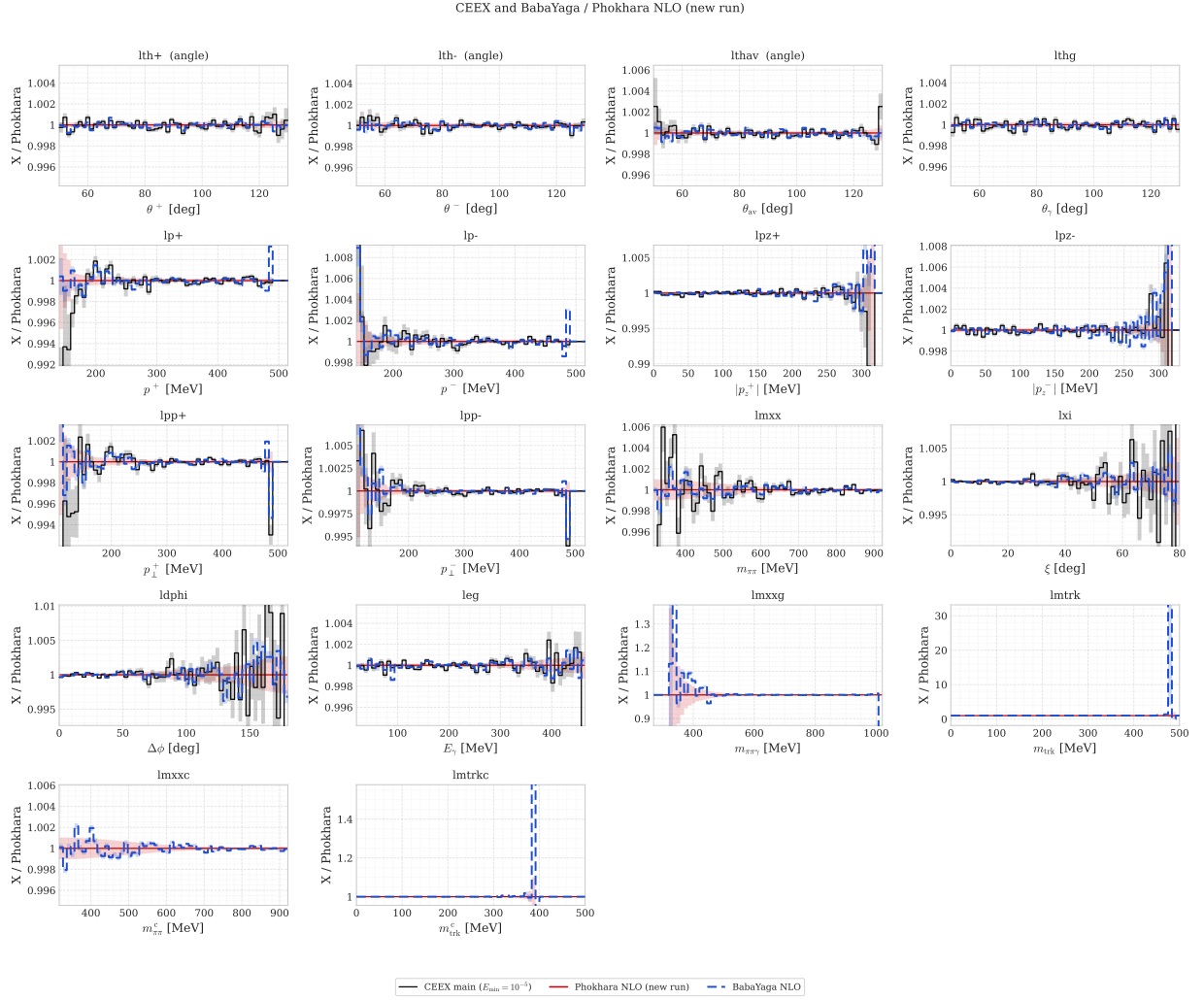


Figure 8: After the fix: every observable, CEEX (more1M, $E_{\min} = 10^{-5}$ GeV) and BabaYaga over the fixed PHOKHARA (1000×5M).

9 Caveats and open points

- Only the configuration studied here was tested ($F_\pi = 1$, no VP, KLOE-LA cuts). Both errors are on code paths active for `pion=1` with `nlo2=1` in any configuration, so they should also affect the Strong2020 form-factor results (there the NLO gap in $A(\theta^+)$ is 0.012: CEEX 0.0804, PHOKHARA 0.0924). This has not been re-run.
- After the fix, one of the three high-statistics samples ($w = 10^{-4}$) lies 0.0007 below the others; an independent sample at the same cutoff agrees with them and with CEEX part by part (Section 8), so it is treated as a fluctuation.
- Muons take a different, correct branch and are not affected.
- Totals and charge-even distributions are not affected by either error (σ unchanged in every run; charge-even per-point agreement 10^{-5}).
- The 0.00027 (2σ) excess of CEEX prod500k over BabaYaga in $A(\theta^+)$ was a statistical fluctuation: CEEX more1M gives $A(\theta^+) = 0.23725(3)$, equal to BabaYaga (0.23724(1)) and the fixed PHOKHARA (0.23725(3)). more1M was still running when this report was written (148 of 200 tasks); the numbers will be updated with the full run.
- With 5×10^9 PHOKHARA events, small BabaYaga–PHOKHARA shape differences become visible in a few charge-even observables (χ^2/ndf 2–3 in $p_\pm$, $p_\pm^\perp$, E_γ ; 10 in $|p_-^z|$, where

BabaYaga’s binning already gave ~ 20 before). CEEX agrees with PHOKHARA there at $\chi^2/\text{ndf} \approx 1$. They are unrelated to the pion-angle issue.

- Tests T5 (charge-class split) and T6 (two-photon point-by-point) of the plan were not needed: the two-photon part agrees (T3) and H1+H2 explain the one-photon part (T4).

10 Other findings

- The checked-out PHOKARA_10.0 has an empty stub `variables_cuts.f` (no cuts, no histograms); the KLOE-I cut file comes from the Strong2020 scenario repository. The card format and the FF_pion numbering changed with respect to the Strong2020 cards.
- PHOKHARA aborts without writing histograms when a weight exceeds the scanned maximum (`phokhara_10.0.f:1910,2135,2263`). Dropping such seeds, as the Strong2020 merge did, biases the result; for weighted events the abort is unnecessary.
- `virt_pions` overwrites its quad-precision momentum arguments (~ 1 GeV). This is harmless in production but breaks any second call in the same event.
- In the 5 Aug reference folder, `Phokhara/NL0novp_*.csv` is the muon result (0.9925 nb), not the pion one.
- CEEX: a few rank outputs per hundred diverge at 100k events (11/300); they are removed before merging. The CEEX-main README still documents `-ref_alloc` and `-vfreeze`, which main no longer parses.
- Cost: at equal precision on σ (prod100k), PHOKHARA needed about 45 times less CPU than CEEX: 0.25 against 28 CPU-hours.

A Reproduction and files

<code>plan.txt</code>	test plan (T0–T6, hypotheses H1–H5)
<code>RUN/config.sh</code>	all paths
<code>RUN/RUN_CEEEX/</code> , <code>RUN/RUN_PHOKHARA/</code>	SLURM submitters and job scripts, runcards
<code>RUN/RUN_PHOKHARA/variants/</code>	<code>h1.patch</code> , <code>h3.patch</code> (the fix), <code>h2.patch</code> (T2 probe), <code>dump.patch</code> (T4), <code>build.sh</code>
<code>RUN/RUN_CEEEX/t4/</code>	CEEX point driver; <code>MC_nphsplit.patch</code> (per-multiplicity histograms)
<code>RUN/POST_PROCESS/</code>	<code>merge</code> , <code>outlier check</code> , <code>phok_table.py</code> , <code>t3_compare.py</code> , <code>t4_compare.py</code>
<code>RESULTS/</code>	<code>prod500k</code> , <code>t1_t2</code> , <code>t3_split</code> , <code>t4_points</code> , <code>t4_fix</code> , <code>fix500k</code> , <code>final500k</code> , <code>fix5M</code> , <code>final5M</code> , <code>final5M_c1M_e5</code> , <code>before_c1M_e5</code>
<code>REPORT/</code>	this report, <code>make_figures.py</code> , <code>make_numbers.py</code>

Building the fixed binary:

```
cd RUN/RUN_PHOKHARA/variants
./build.sh fix h1.patch h3.patch          # -> src/phokhara_fix
cd ../..
EXE=RUN_PHOKHARA/src/phokhara_fix \
  RUN_PHOKHARA/submit.sh <run> <n_seeds> <nges> 10000 <walltime>
```